



King Saud University
College of Computer and Information Sciences
Information Technology department
CSC227: Operating System

Project Report: Operating System

Group#	4	
Section#:	69239	
Group Members	Name	ID
	Mona Alnajjar	444200091
	Rima Alsonbul	444200524
	Jana Alruzuq	444201157
	Remas Al-subaie	444200712
	Razan Aldosari	444201215

Supervised by L. **Rabia Jafri**

Task distribution:

Task distribution	
Name:	Task:
Mona Alnajjar	Contribute in Coding + Test cases
Rima Alsonbul	
Jana Alruzuq	
Remas Al-subaie	
Razan Aldosari	

Screen shots (Input/Output):

The given execution satisfies both conditions:

- 1.FCFS demonstrated:** Since P2 and P3 had the same burst time, P2 executed first as it arrived earlier.
- 2.Preemption demonstrated:** P1 was preempted when P2 arrived with a shorter burst time, and later, P3 was preempted when P5 arrived with an even shorter burst time.

Input:

```
Enter number of processes: 5
Enter arrival time for process 1: 0
Enter burst time for process 1: 7
Enter arrival time for process 2: 1
Enter burst time for process 2: 3
Enter arrival time for process 3: 2
Enter burst time for process 3: 3
Enter arrival time for process 4: 3
Enter burst time for process 4: 5
Enter arrival time for process 5: 4
Enter burst time for process 5: 2
```

Output:

```
Number of processes= 5 (P1, P2, P3, P4, P5)
P1: Arrival time = 0, Burst time = 7 ms
P2: Arrival time = 1, Burst time = 3 ms
P3: Arrival time = 2, Burst time = 3 ms
P4: Arrival time = 3, Burst time = 5 ms
P5: Arrival time = 4, Burst time = 2 ms

Scheduling Algorithm: Shortest Remaining Time First
Context Switch: 1 ms
Time    Process/CS
0-1     P1
1-2     CS
2-5     P2
5-6     CS
6-8     P5
8-9     CS
9-12    P3
12-13   CS
13-18   P4
18-19   CS
19-25   P1

Performance Metrics
Average Turnaround Time: 11.60
Average Waiting Time: 7.60
CPU Utilization: 80.00%
```